DBA5103: Operations Research and Analytics **Semester I, 2026/2027, NUS**

Assignment 2: Due on Sep.27, 2026

1. (10') Find all extreme points in the following polyhedra:

   (a) (2') $P = \{(x_1, x_2, x_3, x_4) \mid x_1 + x_2 + \frac{1}{2}x_3 \leq 1, x_1, x_2, x_3, x_4 \geq 0\}$.

   (b) (4') $P = \{(x_1, x_2, x_3) \mid x_1 - x_2 \leq 1, x_2 - x_3 \leq 1, x_3 \leq 1, x_1, x_2, x_3 \geq 0\}$.

   (c) (4')

$$P = \left\{ (x_1, x_2, x_3, x_4) \;\middle|\; \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, x_1, x_2, x_3, x_4 \geq 0. \right\}$$

2. (10') **Portfolio Management Problem:** Consider the portfolio management problem discussed in class. There are 100 stocks. The expected return of stock i is r_i per dollar invested (which could be negative) and the total budget is b dollars, where $b > 0$. Ignoring the risk of the return, the problem of maximizing the total expected return is

$$\begin{aligned} \max \quad & \sum_{i=1}^{100} r_i x_i \\ \text{s.t.} \quad & \sum_{i=1}^{100} x_i \leq b \\ & x_i \geq 0, \; i = 1, ..., 100, \end{aligned}$$

   where x_i denotes the amount of money invested in stock i.

   (a) Write down the dual problem to the above linear programming problem.

   (b) What is the optimal solution to the primal and the dual respectively?

   (c) Verify the solution you provided in (b) is indeed optimal using the weak duality.

3. (5') Consider the following optimization problem:

$$\begin{aligned} \max \quad & 10x_1 + 12x_2 + 12x_3 \\ \text{s.t.} \quad & x_1 + 2x_2 + 2x_3 \leq 20 \\ & 2x_1 + x_2 + 2x_3 \leq 20 \\ & 2x_1 + 2x_2 + x_3 \leq 20 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

   (a) Write down the dual problem.

   (b) Suppose the optimal solution to the primal problem is $x_1 = x_2 = x_3 = 4$. Solve the optimal solution to the dual problem using complementary slackness condition.

Table 2.1: Profit and resources consumed for each product

	E	**C**	**P_1**	**P_2**	**B**	**Capacity**
Clay (lbs)	10	15	10	10	20	130
Enamel (lbs)	1	2	2	1	1	13
Dry room (hours)	3	1	6	6	3	45
Klin (hours)	2	4	2	5	3	23
Profit	51	102	66	66	89	

4. (10') A potter manufacturer can make four different types of dining room service sets: JJP English, Currier, Primrose, and Bluetail. Furthermore, Primrose can be made by two different methods. Each set uses clay, enamel, dry room time, and kiln time, and results in a profit shown in Table 2.1.

   The manufacturer is currently committed to making the same amount of Primrose using method 1 and method 2. The formulation of the profit maximization problem is given below. The decision variables E, C, P_1, P_2, B are the number of sets of type English, Currier, Primrose Method 1, Primrose Method 2, and Bluetail, respectively. We assume, for the purposes of this problem, that the number of sets of each type can be fractional.

$$\begin{aligned} \max \quad & 51E + 102C + 66P_1 + 66P_2 + 89B \\ s.t. \quad & 10E + 15C + 10P_1 + 10P_2 + 20B \leq 130 \\ & E + 2C + 2P_1 + P_2 + B \leq 13 \\ & 3E + C + 6P_1 + 6P_2 + 3B \leq 45 \\ & 2E + 4C + 2P_1 + 5P_2 + 3B \leq 23 \\ & P_1 - P_2 = 0 \\ & E, C, P_1, P_2, B \geq 0. \end{aligned}$$

   The optimal solution to the primal is given in Table 2.2 and the optimal solution to the dual together with sensitivity information is given in Table 2.3. Use the information to answer the questions that follow.

Table 2.2: The optimal solution to the primal

	Optimal Value	**Objective Coefficient**
E	0	51
C	2	102
P_1	0	66
P_2	0	66
B	5	89

Table 2.3: The optimal solution to the dual and the sensitivity. The last two columns describe the allowed changes in the components of **b** for which the optimal dual solution remains the same

	Dual Variable	**Const. RHS**	**Allowable Increase**	**Allowable Decrease**
Clay	1.429	130	23.33	43.75
Enamel	0	13	∞	4
Dry room	0	45	∞	28
Klin	20.143	23	5.60	3.50
Primrose Methods	11.429	0	3.50	0

(a) What is the optimal quantity of each service set, and what is the total profit?

(b) Give an economic interpretation of the optimal dual variables appearing in the sensitivity report, for each of the five constraints.

(c) Should the manufacturer buy an additional 20 lbs. of Clay at \$1.1 per pound?

(d) In the current model, the number of Primrose produced using method 1 was required to be the same as the number of Primrose produced by method 2. Consider a revision of the model in which this constraint is replaced by the constraint $P_1 - P_2 \geq 0$. In the reformulated problem would the amount of Primrose made by method 1 be positive?

5. (5') **Diet Problem:** The "diet problem" concerns with finding a mix of foods that satisfies requirements on the amount of nutrition. We consider here the problem of choosing fruits to meet certain nutritional requirements. The nutrition information of five selected fruits is provided in Table 2.4 along with a minimum required and maximum allowed intake of each type of nutrition in the last two columns. Your goal is to find the cheapest combination of fruits that will meet nutritional requirements.

Table 2.4: Nutrition and price information per 100 grams

	Apple	**Banana**	**Blueberries**	**Durian**	**Tangerine**	**Min Reqd**	**Max Allowed**
Calories	52	89	57	147	53	500	3000
Carbohydrate (g)	14	23	14	27	13	50	400
Fiber (g)	2.4	2.6	2.4	3.8	1.8	20	30
Vitamin A (IU)	54	64	54	44	681	2000	3500
Vitamin C (mg)	4.6	8.7	9.7	19.7	26.7	75	150
Price (S$/100g)	0.5	0.3	2.5	10	0.5		

(a) Formulate a mathematical model for this problem. Solve the problem using Gurobi and attach your code. Present and interpret the results.

(b) Without solving the problem using the software, can you determine how your optimal solution and optimal spending in (a) will change if you are also concerned with the total fat intake. The fat contained in each of the fruit and your nutrition requirement on fat is provided in the table below. Verify your solution by solving the problem numerically.

	Apple	**Banana**	**Blueberries**	**Durian**	**Tangerine**	**Min Reqd**	**Max Allowed**
Fat (g)	0.2	0.3	0.3	5	0.3	0	10

6. (10') **Markdown Management:** Consider the benchmark model and solution for the markdown management problem discussed in class. The initial inventory is 2000 units. The retailer needs to sell his inventory over a time horizon of 15 weeks by employing in order the four price levels: \$60, \$54, \$48, \$36 over time. The estimated average demands under these price levels are summarized in the following table. Any left over items at the end of the selling season can be sold to outlet for \$25 per

Table 2.5: Benchmark model

Price	\$60	\$54	\$48	\$36
Demand	125	162.5	217.5	348.8

unit (salvage value). The retailer is required to start with the full price for at least one week and then he is allowed to decrease his prices over time. We have shown in class the optimal policy is to use for the first 11.67 weeks of full price \$60 and then 3.33 weeks of \$54 (ignoring integrality constraints). The revenue collected under this policy is \$116750. Consider the following three scenarios **separately** and answer the corresponding questions. Attach the codes you used to derive the solutions.

(a) Suppose in order to maintain the brand reputation, the retailer is only allowed to discount from week 4. Will the retailer change his markdown strategy in this case?

(b) Due to bad weather, only 25 units were sold in the first week. How should the retailer adjust his markdown policy? What is the improvement in revenue compared to the case when he sticks with the original markdown policy (specified above)?

(c) In the traditional brick-and-mortar business, menu cost and consumer price impression are the two major reasons behind the very few number of price levels. With the rise of e-commerce and other technologies, it is possible for the retailer to employ more price levels and change the price more frequently. What happens if the retailer employs the following price grids (with a grid of \$2) instead of a few discount levels? For simplicity, the demand estimates at the additional price levels are taken as a linear interpolation of the original demands. What is the improvement in revenue compared to the benchmark? What is the new markdown policy? How do you compare with the original policy?

Table 2.6: Price grids model

Price	\$60	\$58	\$56	\$54	\$52	\$50	\$48	\$46	\$44	\$42	\$40	\$38	\$36
Demand	125	137.5	150	162.5	180.8	199.1	217.5	239.4	261.3	283.2	305.1	327	348.8